

The 4-tuples in a 4-ary relation represent these attributes of published books: title, ISBN, publication date, number of pages.

- a) What is a likely primary key for this relation?
- b) Under what conditions would (title, publication date) be a composite key?
- c) Under what conditions would (title, number of pages) be a composite key?

Show that if C_1 and C_2 are conditions that elements of the n -ary relation R may satisfy, then $s_{C_1 \wedge C_2}(R) = s_{C_1}(s_{C_2}(R))$.